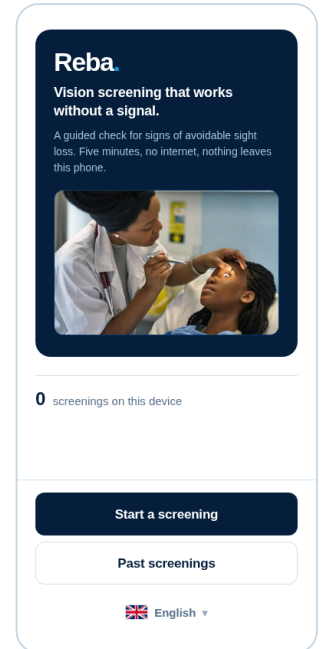




Vision screening for community health workers in Rwanda.

Five minutes. One cheap phone. No network.



THE PROBLEM

Two treatable causes of childhood blindness

- **Retinoblastoma.** A cancer of the eye. Found early, the child lives. Missed, the child dies.
- **Amblyopia.** An eye that never learns to see. By age seven the loss is permanent.

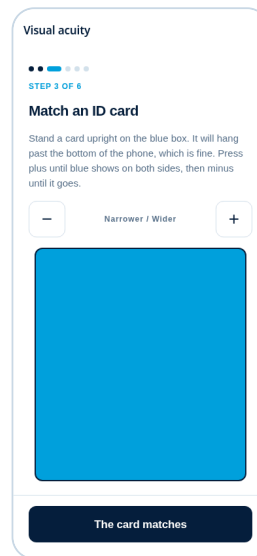
Neither one hurts. The child never complains.

The person who sees the child first is a community health worker, and they have nothing to look with.

ONE SCREENING

Five minutes, start to finish

- 1 **Consent**, read aloud in the family's language
- 2 **Vision test**, calibrated against an ID card so the letters are a true size
- 3 **One photograph of each eye**, flash on
- 4 **A result**: refer, recheck, or no signs today
- 5 **A referral slip** to hand to the family



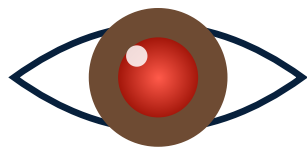
The card sets the scale



A true size letter

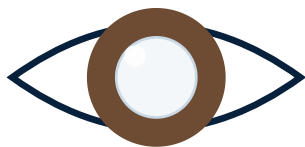
MEASURED, ON THE DEVICE

The red reflex, compared between the two eyes



Red

the path is clear



White

something is in the way

CLINICAL
PHOTOGRAPHS

HEALTHY
EYE

AFFECTED
EYE

Child 1

0.475

0.381

Child 2

0.476

0.375

Both affected eyes were flagged. Neither healthy eye was. Same phone, same flash, seconds apart, so skin tone and lighting cancel out. **The patient is their own control.**

HONESTY

What is real, and what is not

WORKING TODAY

- Consent and records
- Calibrated visual acuity
- Eye photographs, kept on the phone
- Triage and referral

NOT YET

- A trained model
- Thresholds validated on real eyes
- Kinyarwanda checked by a native speaker

A model needs thousands of eye photographs labelled by ophthalmologists and taken on phones in villages. That dataset does not exist for Rwanda, because nobody had the tool to collect it.

Reba is that tool.

WHAT I AM ASKING FOR

Three things

- **A clinical partner** to label the photographs the app is already collecting
- **A pilot** with community health workers in one district
- **A Kinyarwanda speaker** to review the consent script before any real patient